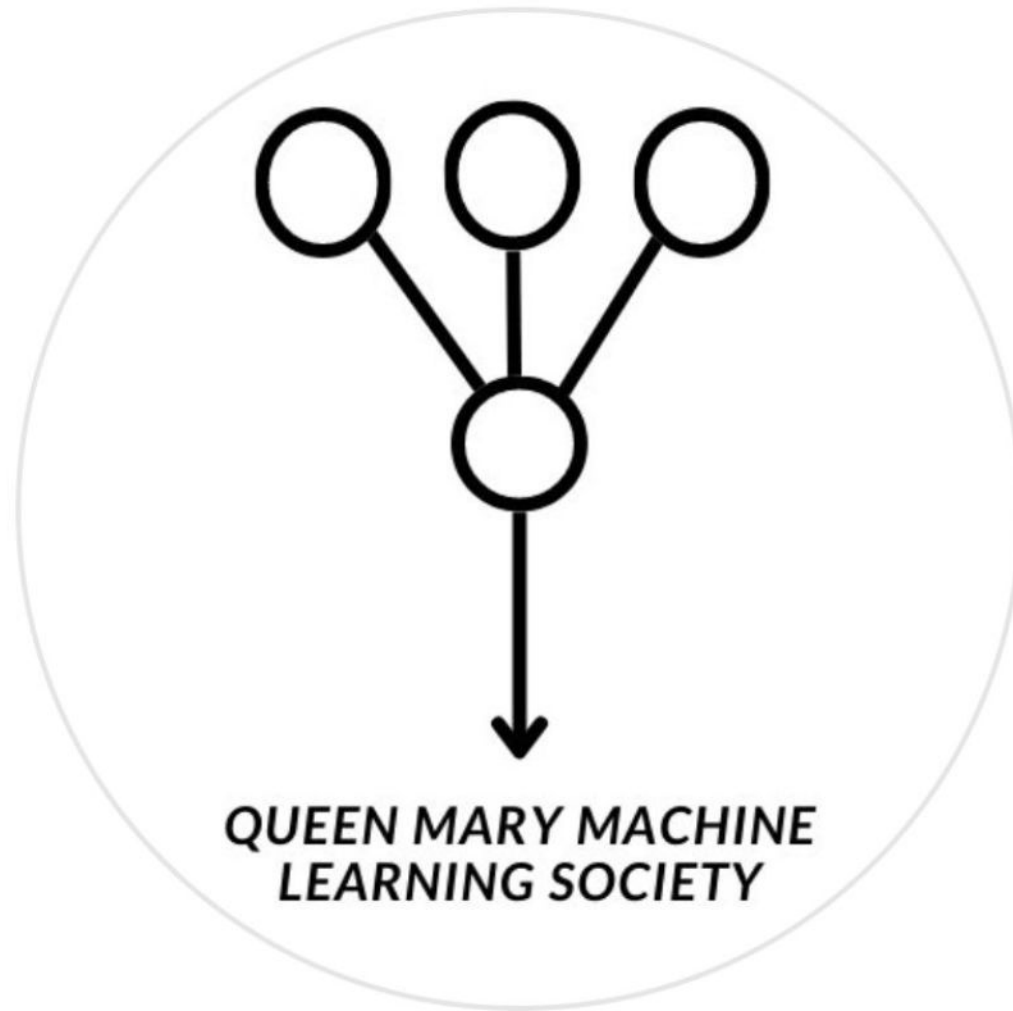
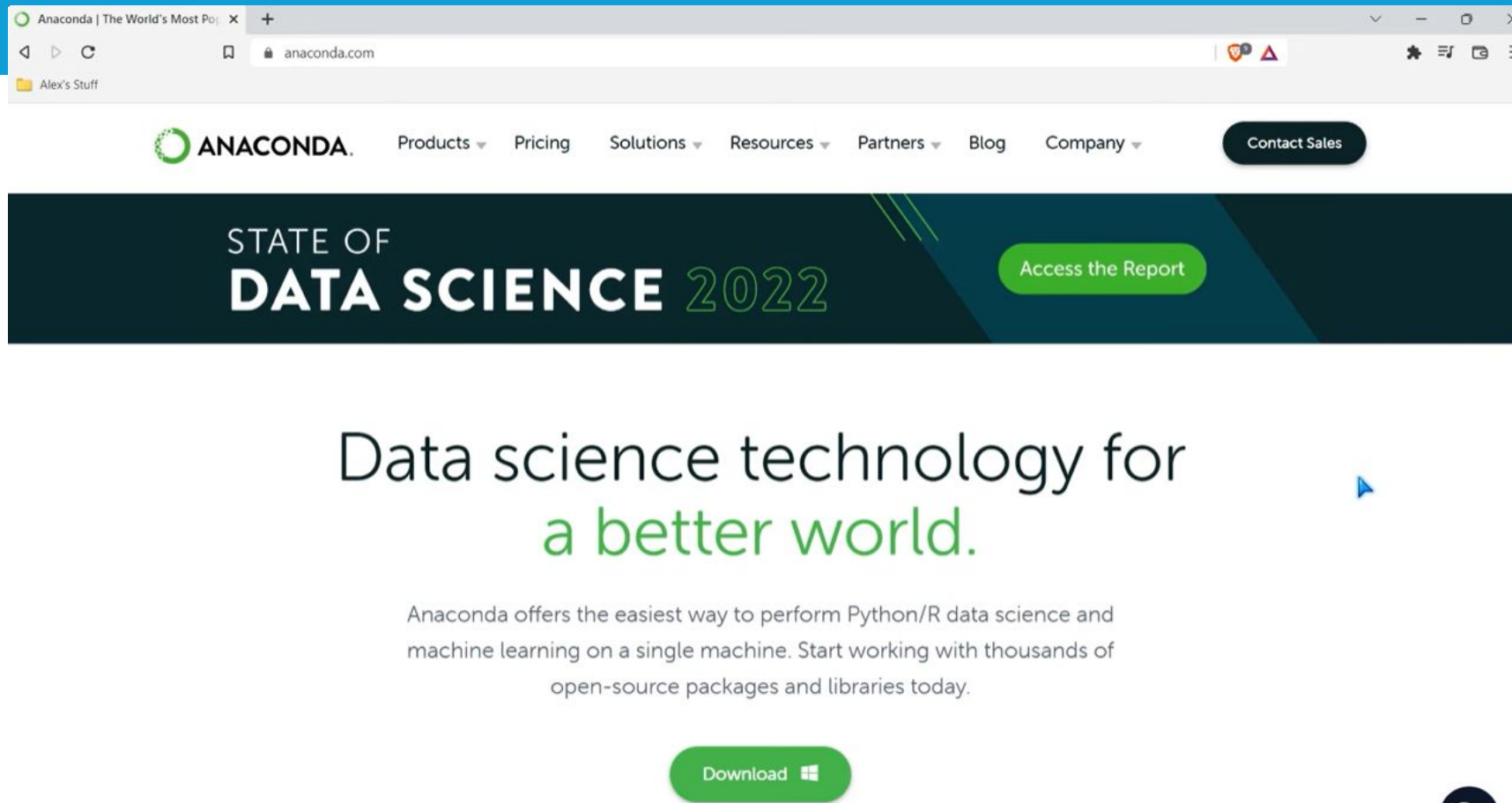


Kaggle #02



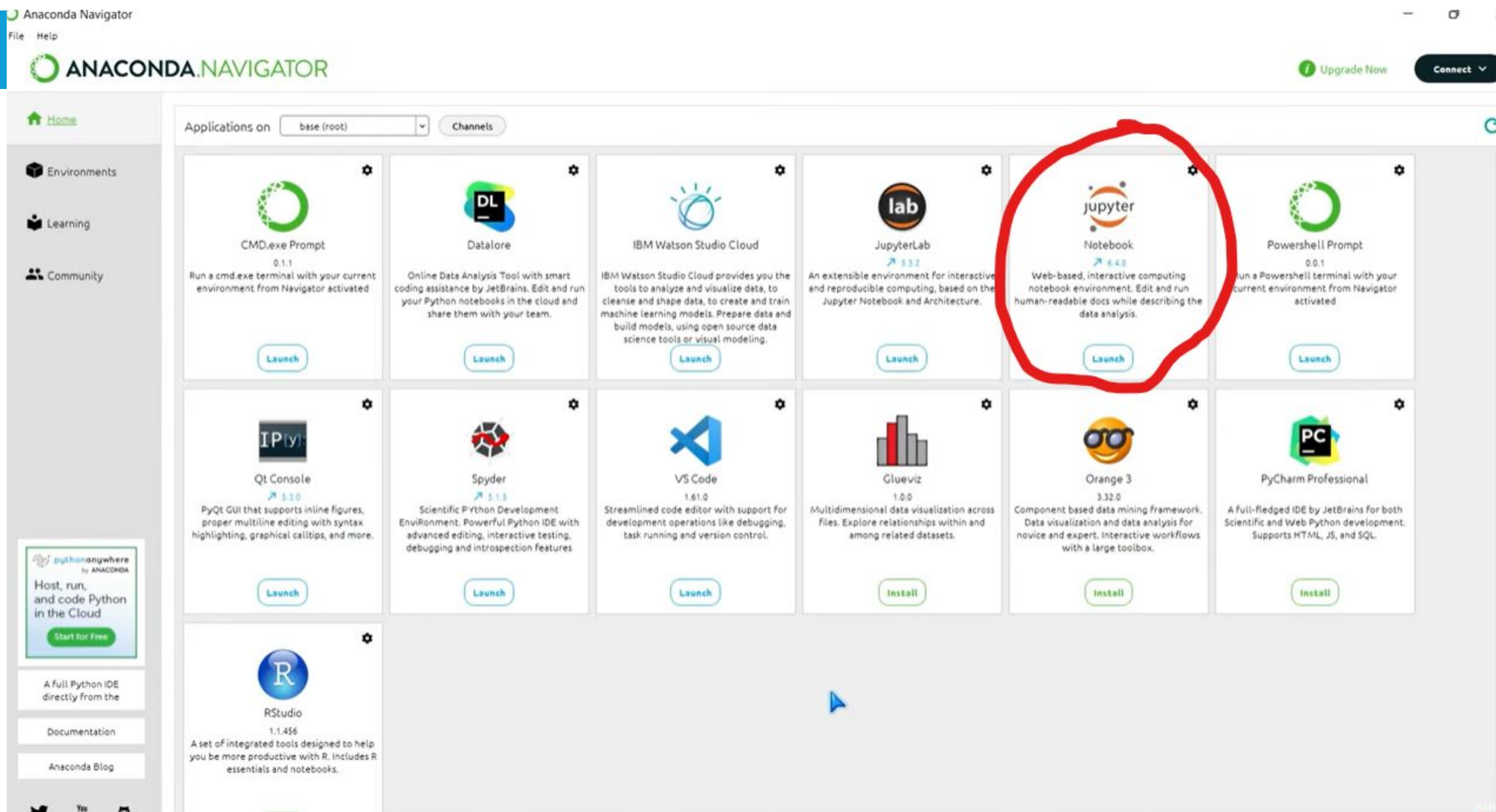
Downloading Anaconda and Jupyter notebook



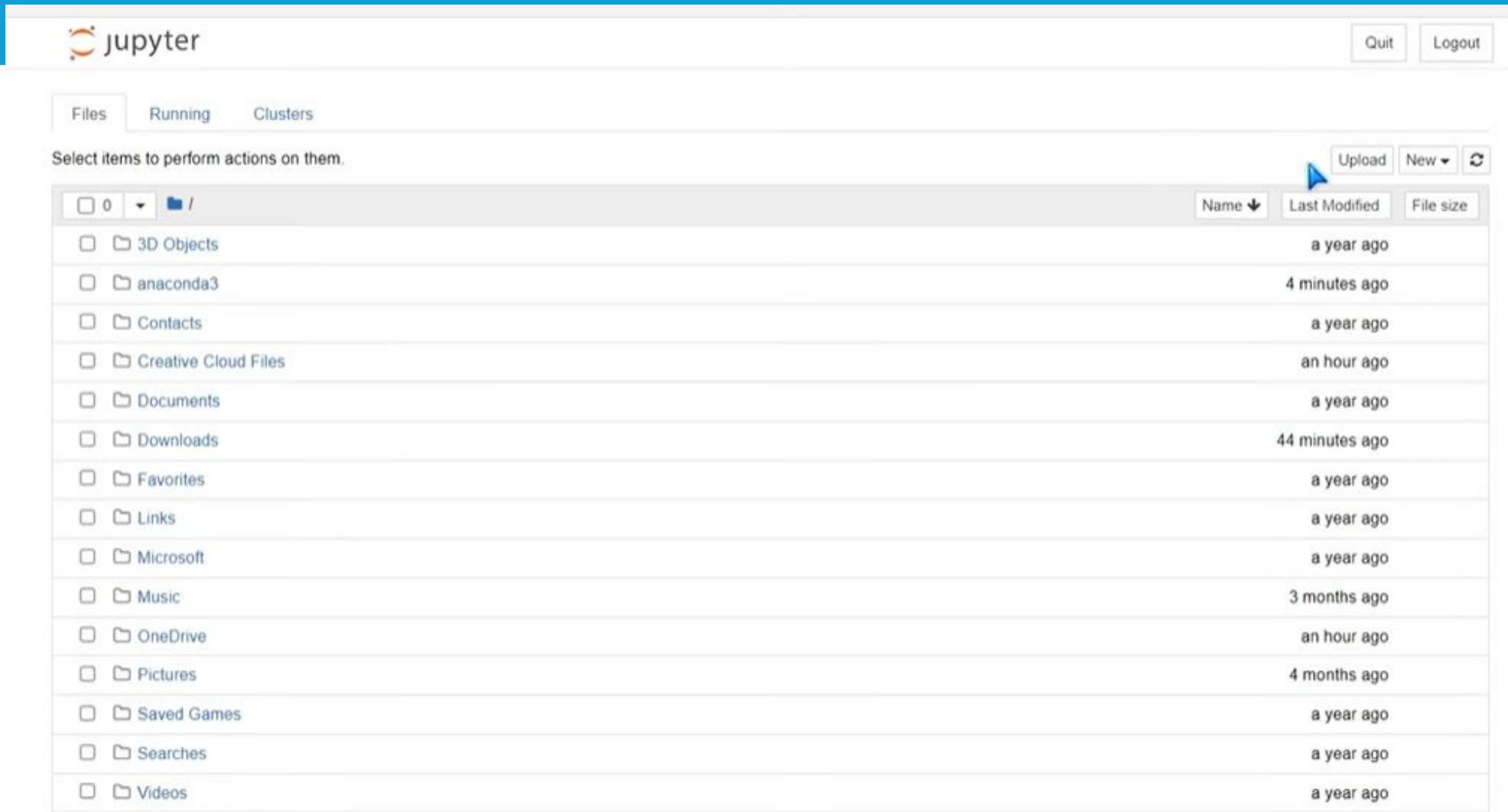
Follow Alex Analyst's tutorial :

<https://www.youtube.com/watch?v=WUeBzT43JyY&t=232s>

Downloading Anaconda and Jupyter notebook



Downloading Anaconda and Jupyter notebook



The screenshot displays the JupyterLab web interface. At the top, the 'jupyter' logo is on the left, and 'Quit' and 'Logout' buttons are on the right. Below the logo, there are tabs for 'Files', 'Running', and 'Clusters'. The 'Files' tab is active, showing a message 'Select items to perform actions on them.' and a toolbar with 'Upload', 'New', and a refresh icon. A blue arrow cursor points to the 'New' button. Below the toolbar is a table listing files and folders in the root directory ('/'). The table has columns for 'Name', 'Last Modified', and 'File size'. The list includes folders like '3D Objects', 'anaconda3', 'Contacts', 'Creative Cloud Files', 'Documents', 'Downloads', 'Favorites', 'Links', 'Microsoft', 'Music', 'OneDrive', 'Pictures', 'Saved Games', 'Searches', and 'Videos'. Each item has a checkbox on the left and a timestamp on the right.

☐ 0	Name	Last Modified	File size
☐	3D Objects	a year ago	
☐	anaconda3	4 minutes ago	
☐	Contacts	a year ago	
☐	Creative Cloud Files	an hour ago	
☐	Documents	a year ago	
☐	Downloads	44 minutes ago	
☐	Favorites	a year ago	
☐	Links	a year ago	
☐	Microsoft	a year ago	
☐	Music	3 months ago	
☐	OneDrive	an hour ago	
☐	Pictures	4 months ago	
☐	Saved Games	a year ago	
☐	Searches	a year ago	
☐	Videos	a year ago	

What is Pandas

- Library in python - lets you work with tabular data (like excel/ google sheets)
- Python = general purpose language you use to communicate with the computer like English is for people



Analogy:

Python = the kitchen



Pandas = a tool like a set of knives, spoons, and cutting boards that make chopping, slicing, and preparing ingredients (data) much easier

Pandas Concepts

Series = column in a table

Dataframe = 2D data structure like a table

```
import pandas as pd
```

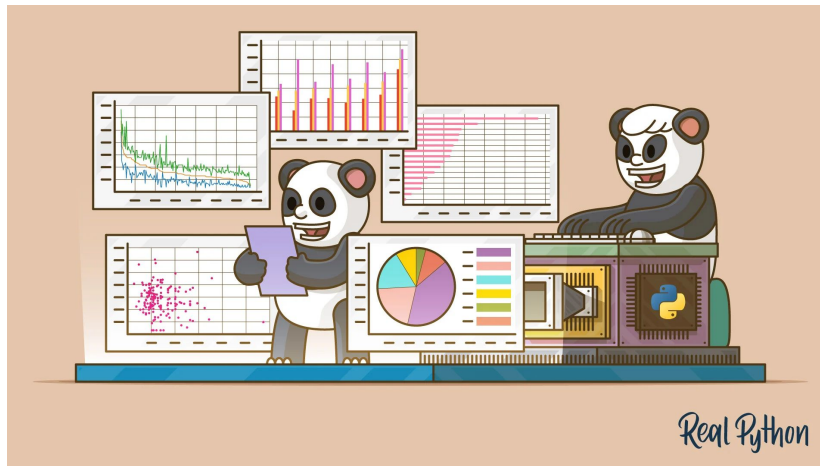
```
data = {  
    "calories": [420, 380, 390],  
    "duration": [50, 40, 45]  
}
```

```
df = pd.DataFrame(data, index = ["day1", "day2", "day3"])
```

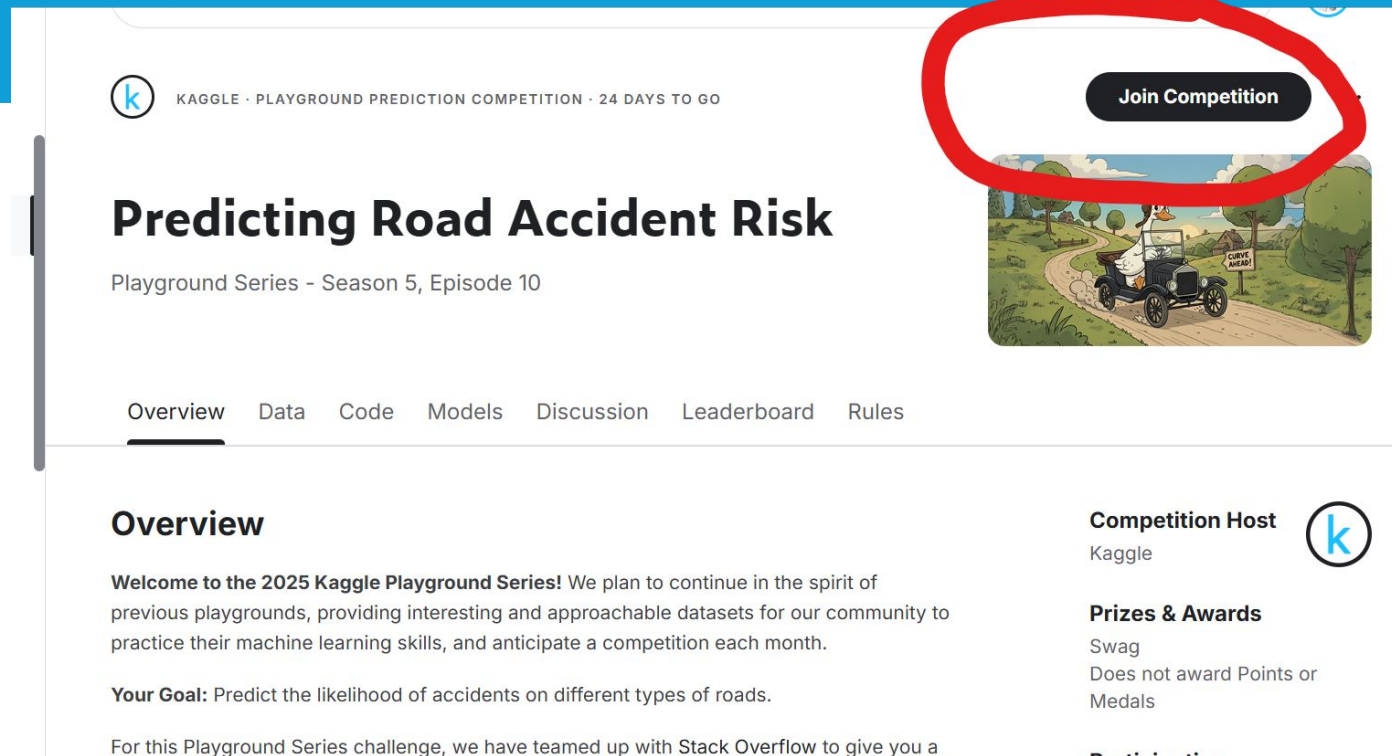
```
print(df)
```

Pandas Syntax Basics

- Has functions for analyzing, cleaning, exploring, and manipulating data.
- `Import pandas` = imports the library
- `pd.read_csv()` = reads in file
- `pd.DataFrame()` = convert to pandas data structure
- `df['columnName']` = returns the column
- `df['columnName'].mean()` = return mean of the column



Recap: Playground Series



The screenshot shows the Kaggle Playground Prediction Competition page for 'Predicting Road Accident Risk'. The page is titled 'KAGGLE · PLAYGROUND PREDICTION COMPETITION · 24 DAYS TO GO'. The main heading is 'Predicting Road Accident Risk', followed by 'Playground Series - Season 5, Episode 10'. A red circle highlights the 'Join Competition' button. Below the title is a navigation bar with links: Overview, Data, Code, Models, Discussion, Leaderboard, and Rules. The 'Overview' section is active. The 'Overview' text reads: 'Welcome to the 2025 Kaggle Playground Series! We plan to continue in the spirit of previous playgrounds, providing interesting and approachable datasets for our community to practice their machine learning skills, and anticipate a competition each month.' The 'Your Goal' is to 'Predict the likelihood of accidents on different types of roads.' The page also mentions a partnership with Stack Overflow. On the right, the 'Competition Host' is Kaggle, and the 'Prizes & Awards' section states 'Swag' and 'Does not award Points or Medals'.

KAGGLE · PLAYGROUND PREDICTION COMPETITION · 24 DAYS TO GO

Predicting Road Accident Risk

Playground Series - Season 5, Episode 10

[Join Competition](#)

[Overview](#) [Data](#) [Code](#) [Models](#) [Discussion](#) [Leaderboard](#) [Rules](#)

Overview

Welcome to the 2025 Kaggle Playground Series! We plan to continue in the spirit of previous playgrounds, providing interesting and approachable datasets for our community to practice their machine learning skills, and anticipate a competition each month.

Your Goal: Predict the likelihood of accidents on different types of roads.

For this Playground Series challenge, we have teamed up with [Stack Overflow](#) to give you a

Competition Host 
Kaggle

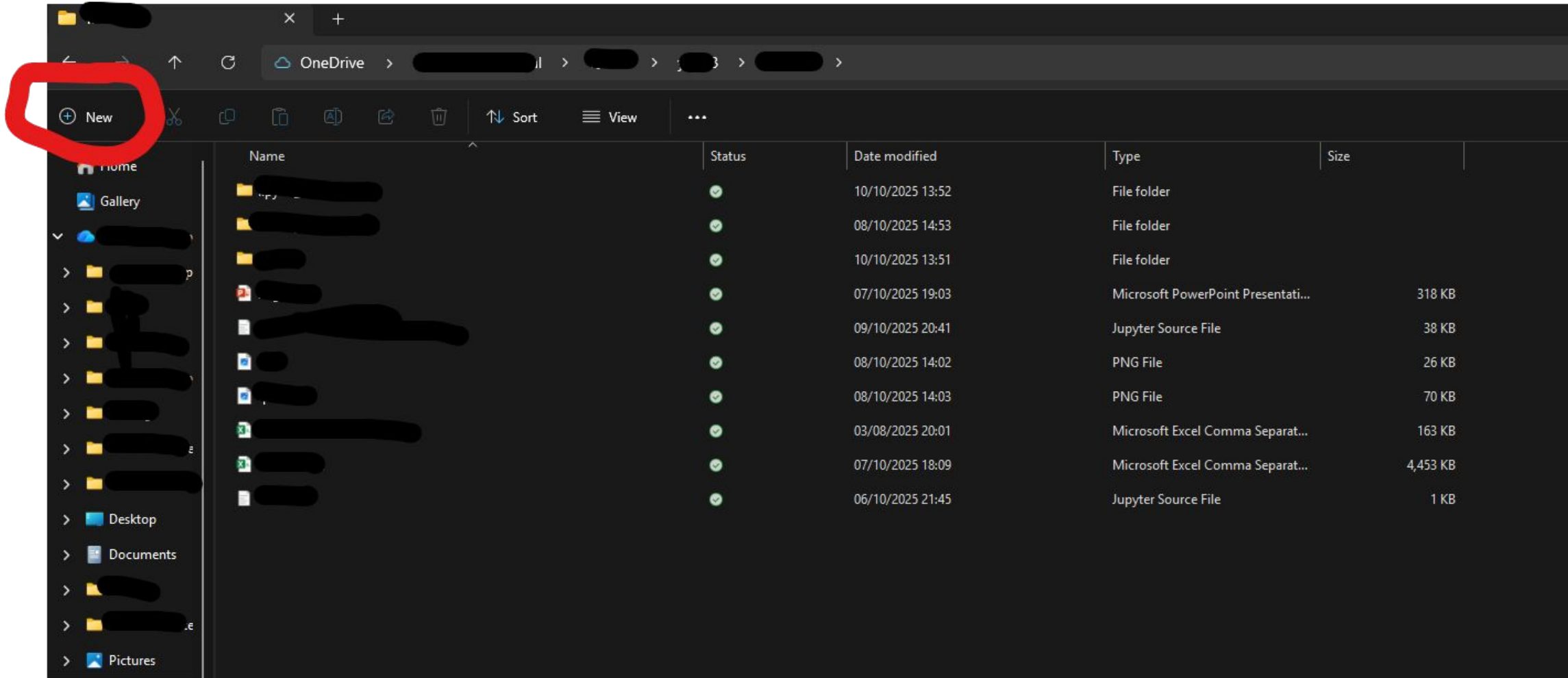
Prizes & Awards
Swag
Does not award Points or Medals

Link to dataset: <https://www.kaggle.com/competitions/playground-series-s5e10/overview>
Due: October 31 2025 11:59 PM UTC

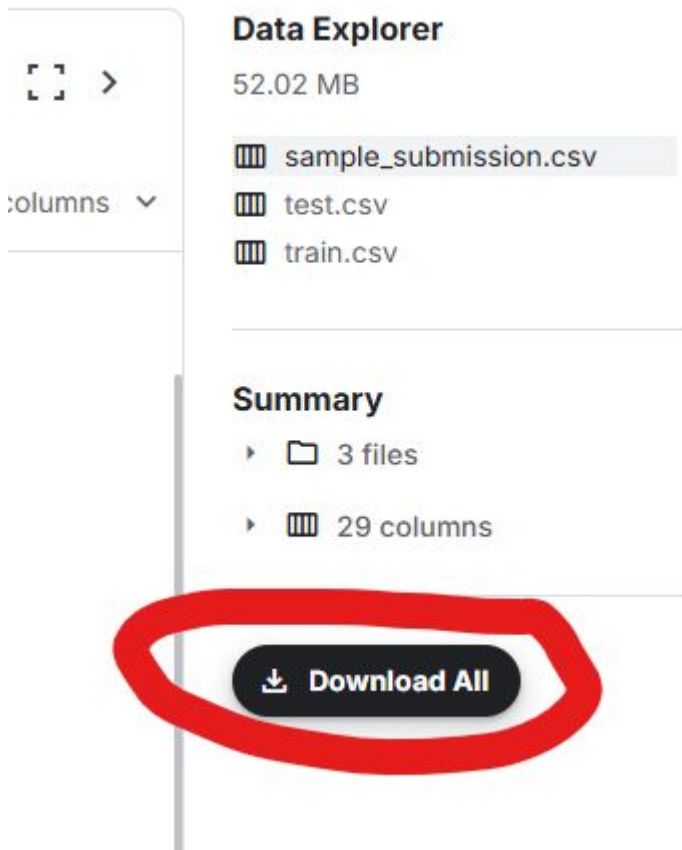
Part 2: Hands on Tutorial



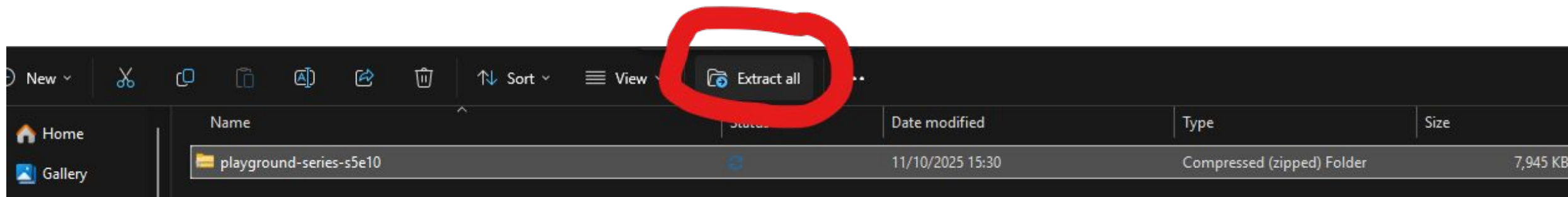
Go to Files and create a Folder



Maybe call it Kaggle or Kaggle Projects so it's easy to find



1. On the Kaggle website, click download all and save it to the new Folder we created in the previous step
2. Extract all the files in “Kaggle Projects” folder
3. You can go ahead and delete the old playground series folder



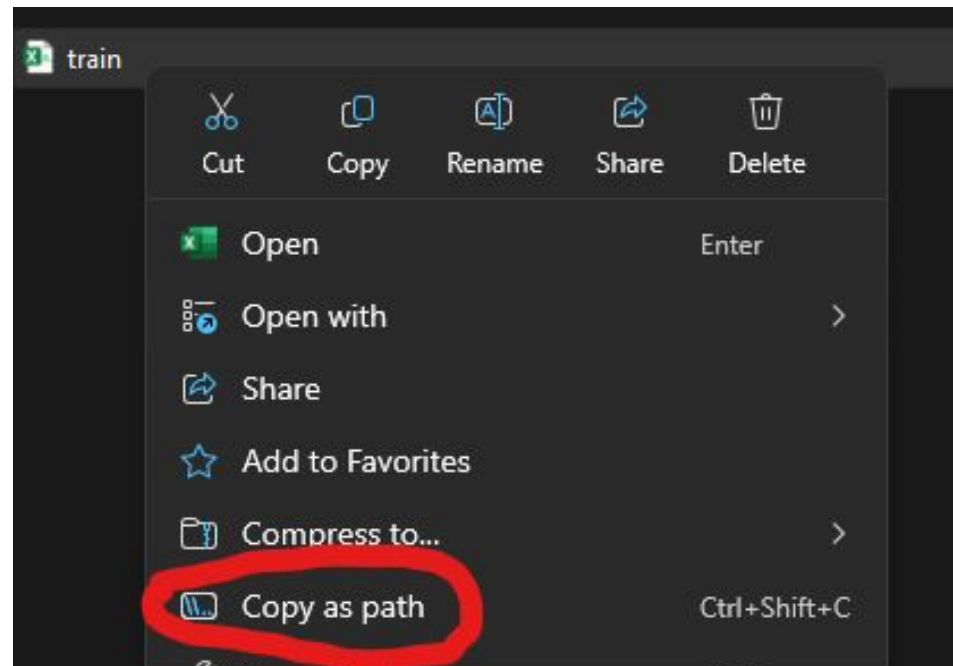
Download Template or use your own!

1. Go to QMML github -> then 'Kaggle Seasons' and download the sample template to "Kaggle Project" folder on your computer
2. Open Jupyter Notebook -> Navigate to "Kaggle Folder" and select this sample template notebook (.ipynb file) and open it
3. We can start coding!

Github link: <https://github.com/Queen-Mary-Machine-Learning-Society/QMML>

1. Importing Relevant libraries and reading in data

- `pd.read_csv(.....)` - read in data
- To get the file path easily:
 - Right click on a dataset then select copy as path



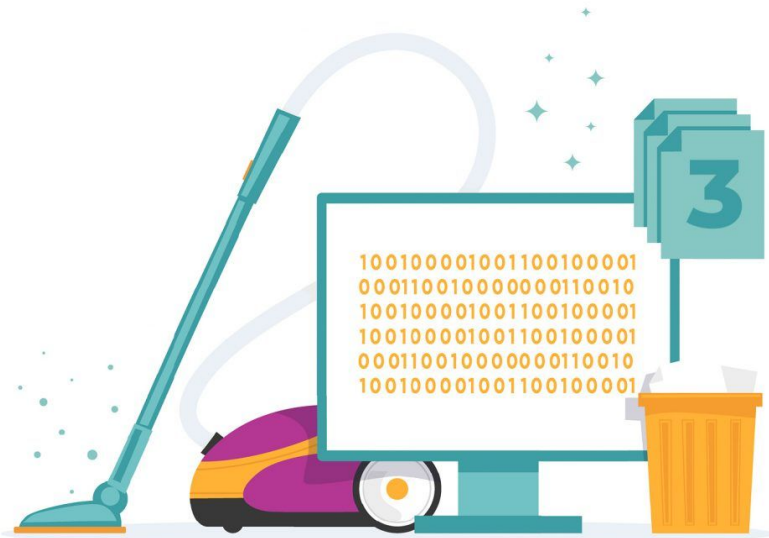
2. EDA

- `df.head()` = shows first 5 rows
- `Df.columns` = shows all the names of the columns
- `Df.shape` = returns the no. rows and columns
- `Df.describe` = returns statistics about the data like mean etc.



3. Data Cleaning

- `df.isnull().sum()` = checks the number of NaN (missing) values
- `df.drop(columns = ..., axis = 1)` = drops specified columns from dataset
- `df.dropna()` = drops missing values
- `df.fillna()`



4. Encoding

- **What it is:** Converting categorical or text data into numerical form so ML models can process it.
- **Why it matters:** Models work with numbers, not words or labels.
- **Types of Encoding:**
 - **Label Encoding** - Assigns each category a number (e.g., Red=0, Blue=1, Green=2).
 - **One-Hot Encoding** - Creates binary columns (e.g., Red = [1,0,0], Blue = [0,1,0], Green = [0,0,1]).
 - **Ordinal Encoding** - For categories with natural order (e.g., Small=1, Medium=2, Large=3).
- The right encoding depends on the type of data and model being used

Matplotlib

Creates high quality charts
for analysis and
publication.

Core library for data
visualisation in ML!



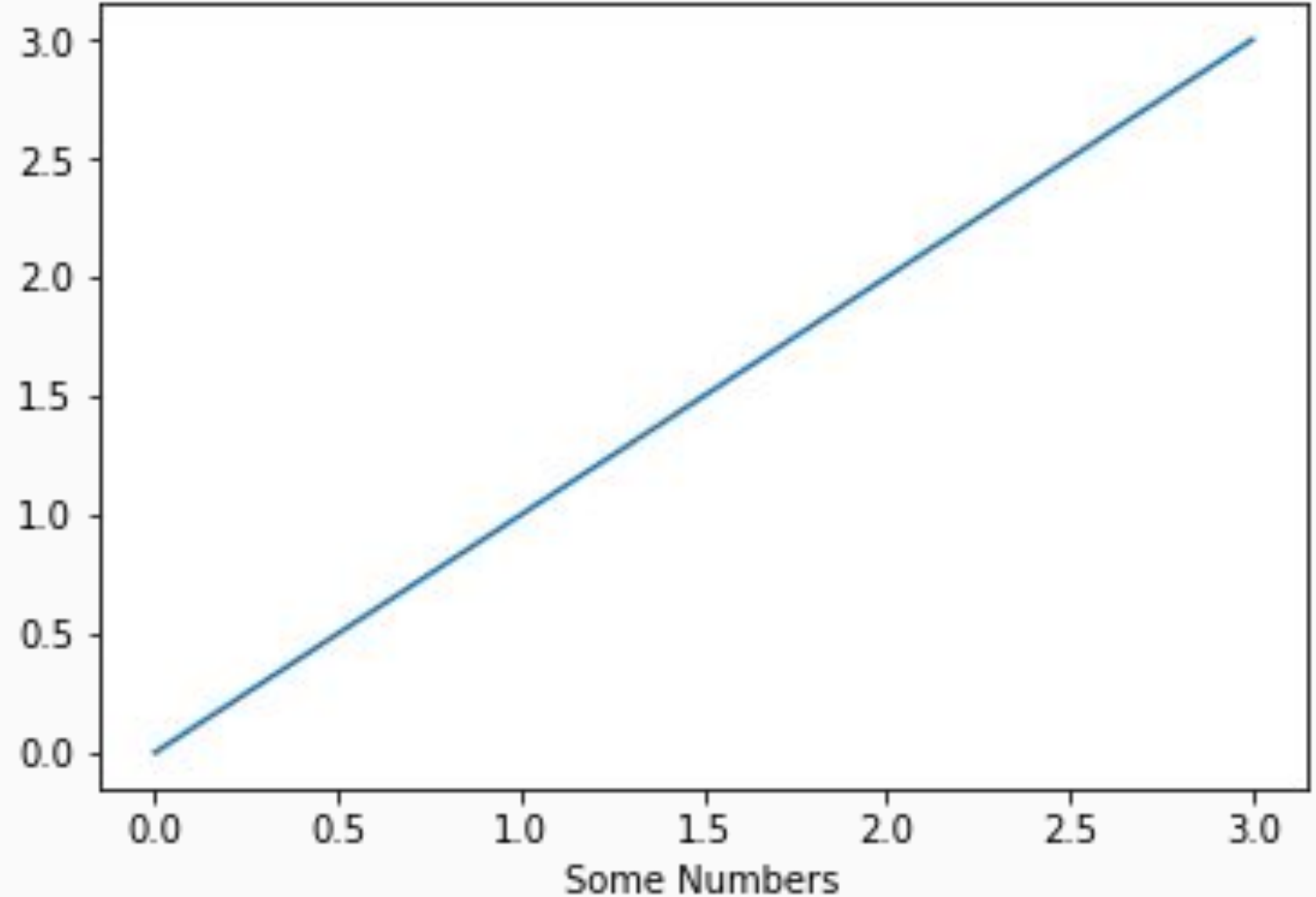
Firstly, Import!

```
import matplotlib.pyplot as plt
```

Basic Example

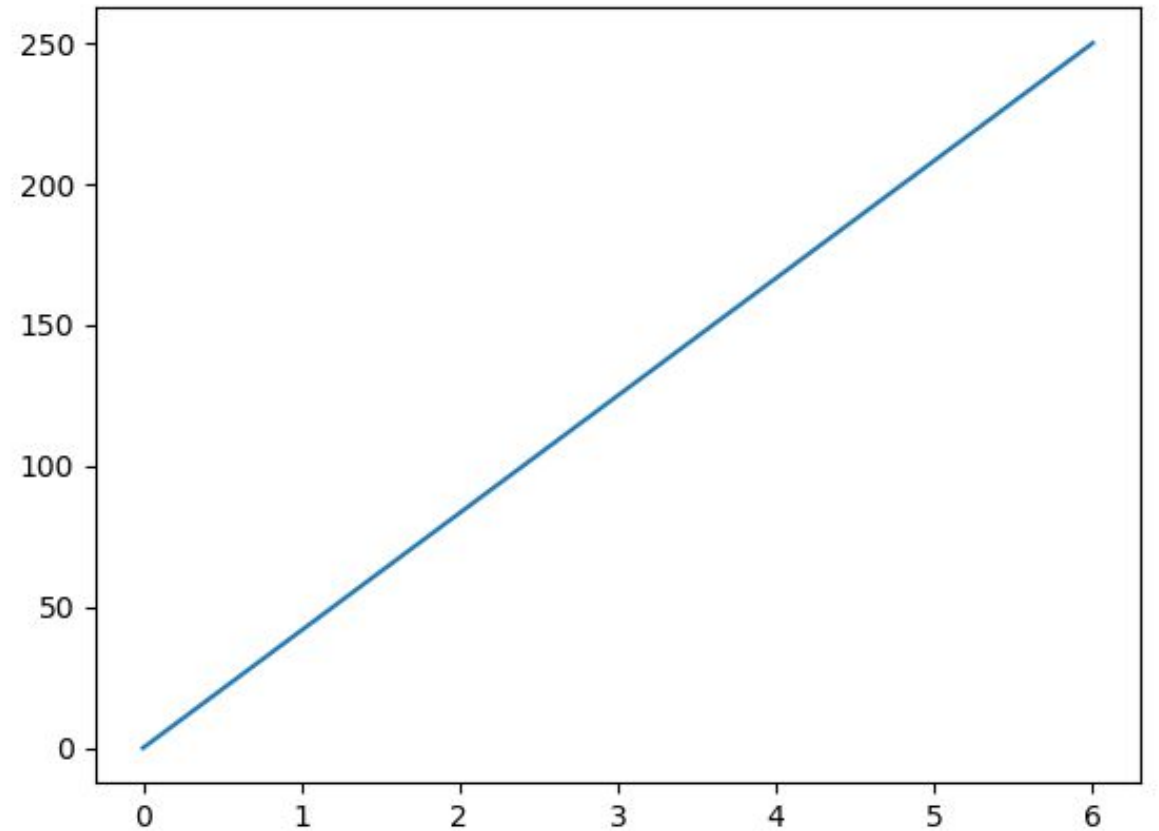
```
plt.plot([0, 1, 2, 3])
```

```
plt.xlabel("Some  
Numbers");
```



Another Example

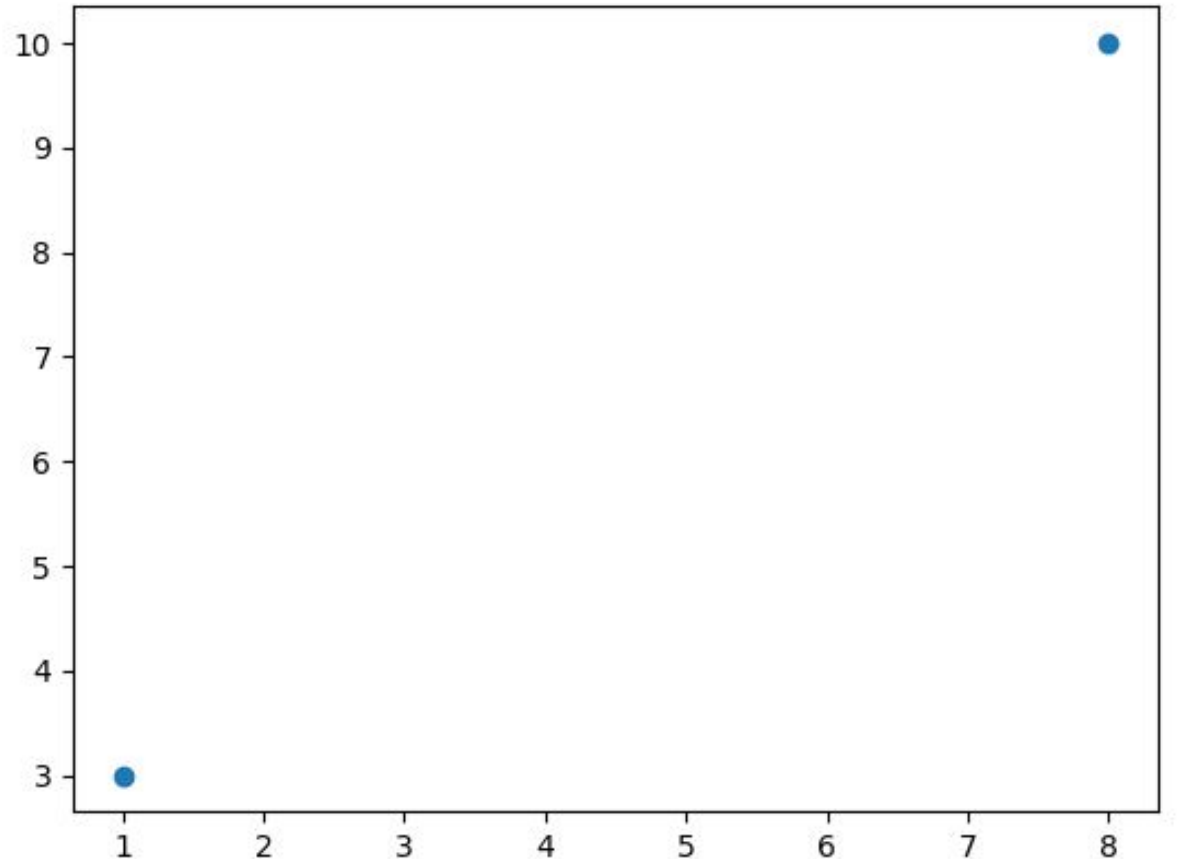
```
xpoints = np.array([0, 6])  
ypoints = np.array([0, 250])  
  
plt.plot(xpoints, ypoints)
```



Plot without Line

Just add an 'o'

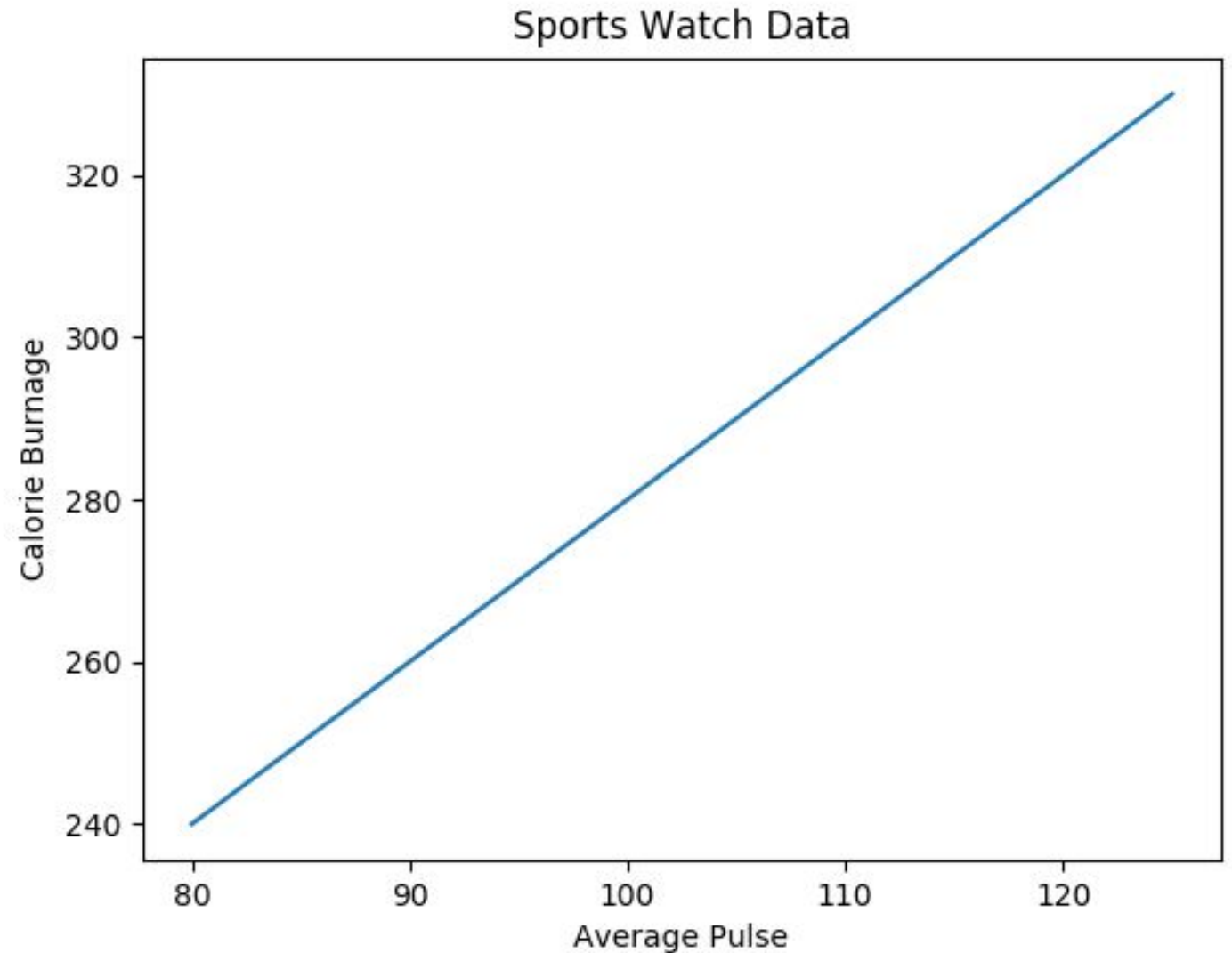
```
xpoints = np.array([1, 8])  
ypoints = np.array([3, 10])  
  
plt.plot(xpoints, ypoints,  
         'o')
```



Labels

Title, xlabel, ylabel

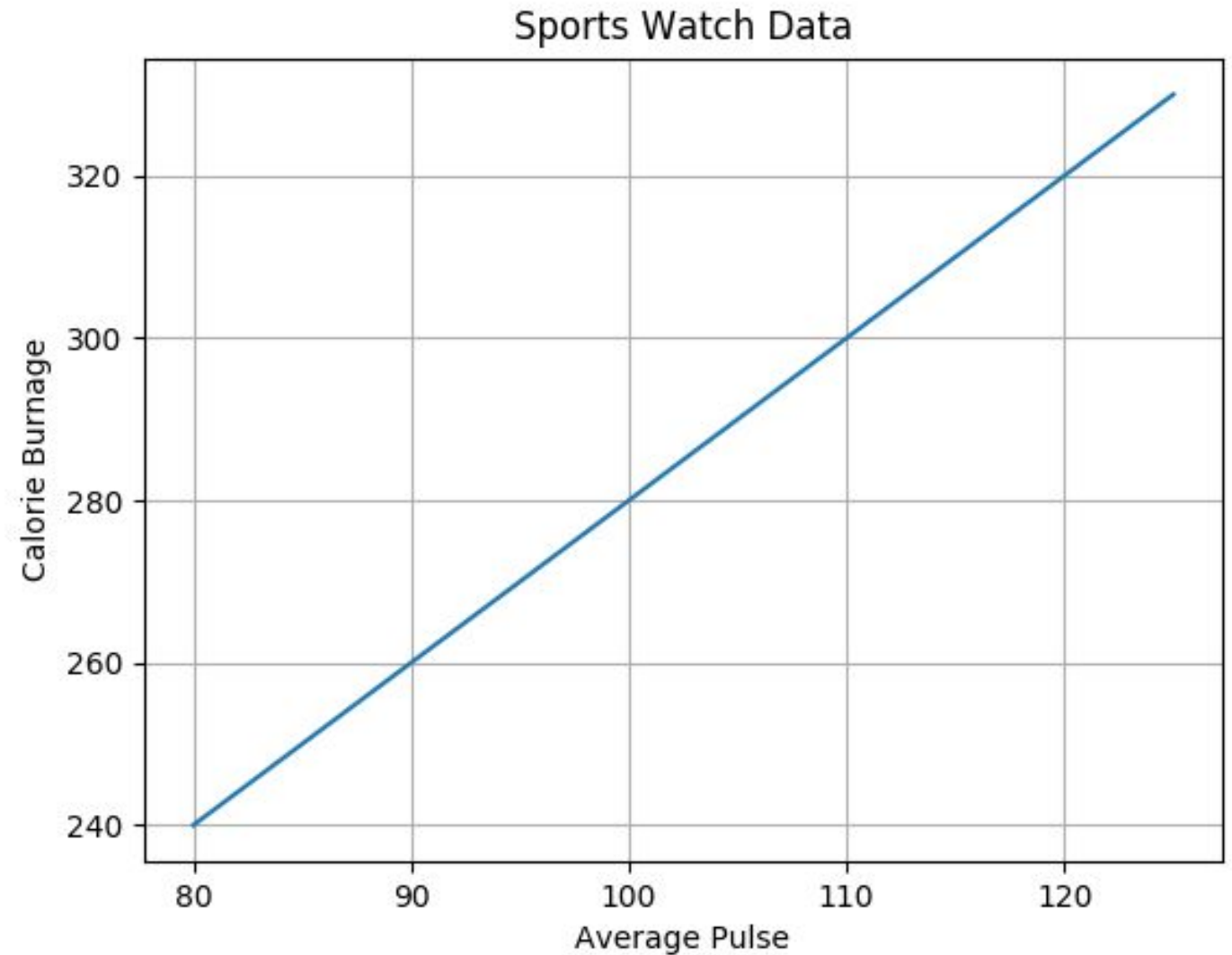
```
plt.title("Sports Watch Data")  
plt.xlabel("Average Pulse")  
plt.ylabel("Calorie Burnage")
```



Add Grids!

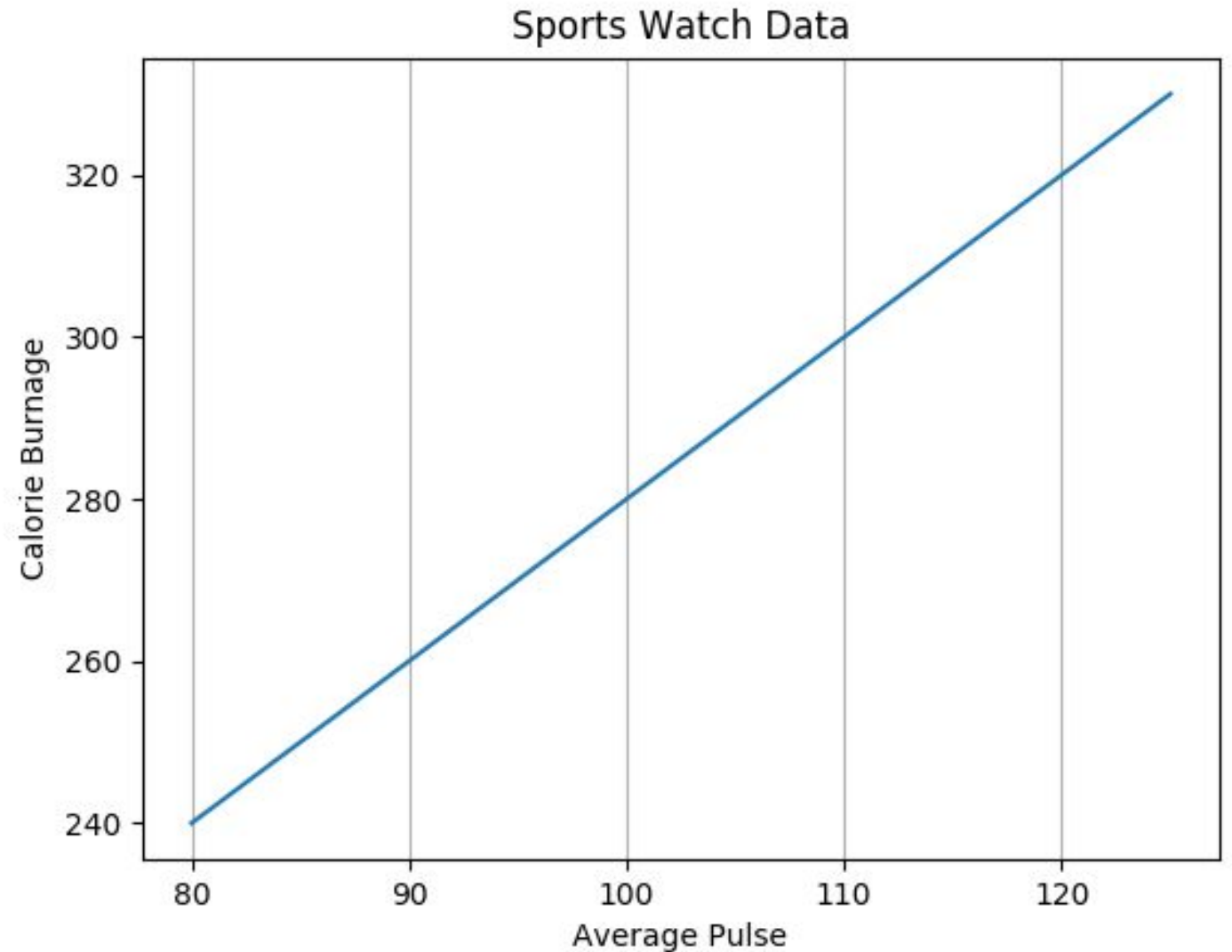
with `plt.grid()`

`plt.grid()`



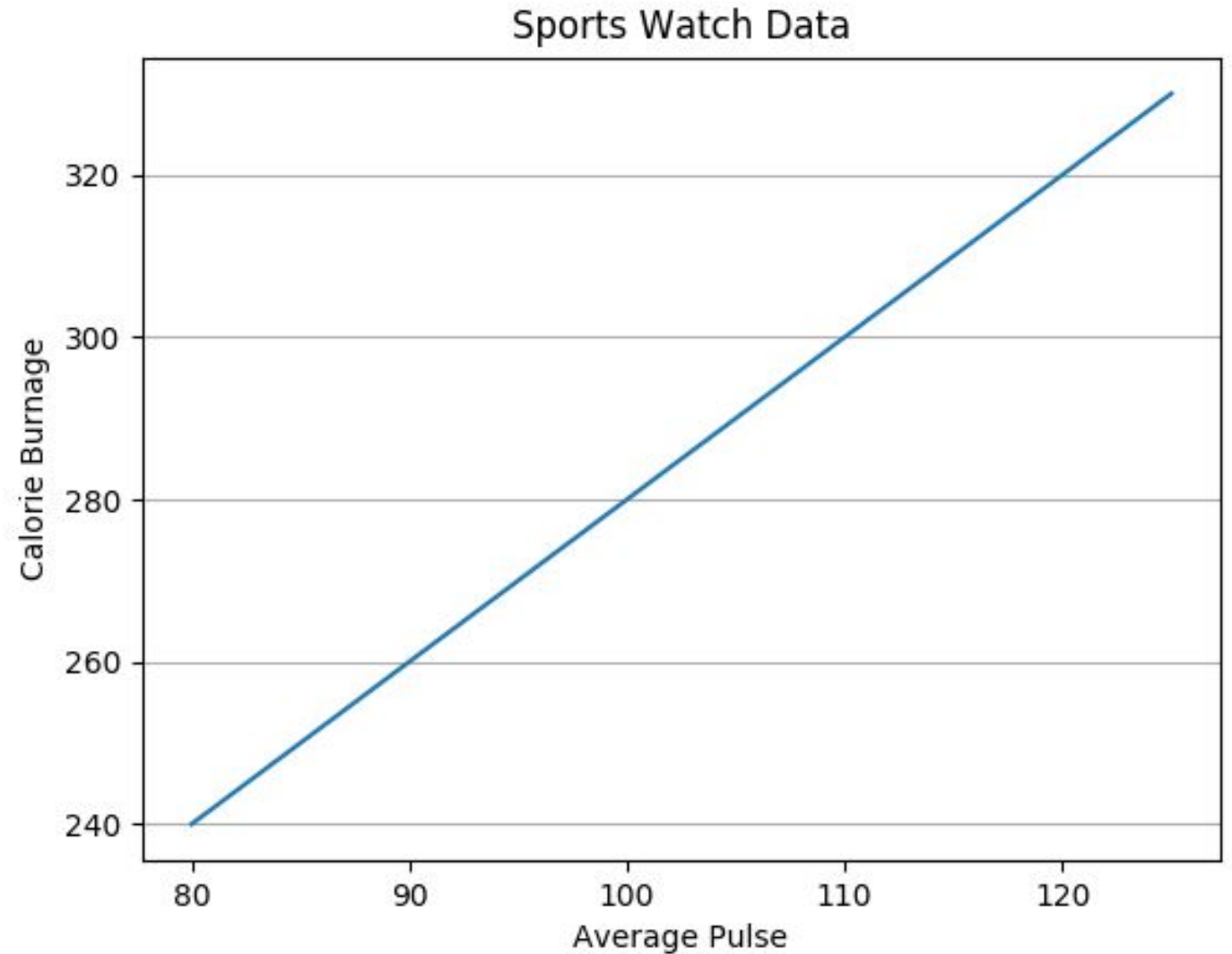
Specify Axis

```
plt.grid(axis = 'x')
```



Specify Axis

```
plt.grid(axis = 'y')
```



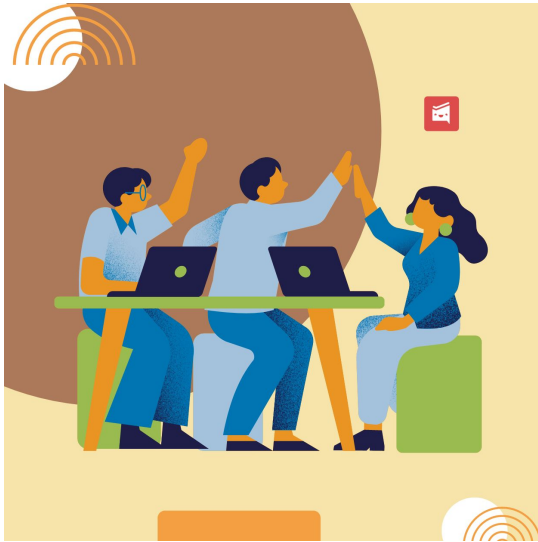
What can matplotlib plot?

Line Plot	<code>plt.plot()</code>
Scatter Plot	<code>plt.scatter()</code>
Bar Chart	<code>plt.bar()</code>
Histogram	<code>plt.hist()</code>
Box Plot	<code>plt.boxplot()</code>
Advanced	<code>plt.figure(),</code> <code>plt.subplots()</code>

Some real life examples!

Let's Get into Teams!

- Open the spreadsheet from the QR link
- Pick a row and Enter your Team name + Your names
- Get into teams of 3 people
- Come up if you can't find a team! We will group you up.



https://docs.google.com/spreadsheets/d/1HNIhKU8CkD-5e0b14W2KVPZ2ItB7bTK2hT_NKgNyQxl/edit?usp=drivesdk

Kaggle Feedback

<https://forms.gle/QL4hqfEHYjAZZGiV9>

